

Calculating and Interpreting Correlation Coefficients

Is Animal Complexity Correlated with miRNA Diversity? Animal phyla vary greatly in morphology, from simple sponges that lack tissues and symmetry to complex vertebrates. Members of different animal phyla have similar developmental genes, but the number of miRNAs varies considerably. In this exercise, you will explore whether miRNA diversity is correlated to morphological complexity.

How the Study Was Done In the analysis, miRNA diversity is represented by the average number of miRNAs in a phylum (x), while morphological complexity is represented by the average number of cell types (y). Researchers examined the relationship between these two variables by calculating the correlation coefficient (r). The correlation coefficient indicates the extent and direction of a linear relationship between two variables (x and y) and ranges in value between -1 and 1 . When $r < 0$, x and y are negatively correlated, meaning that in a plot of the data points, values of y become smaller as values of x become larger. When $r > 0$, x and y are positively correlated (y becomes larger as x becomes larger). When $r = 0$, the variables are not correlated.

The formula for the correlation coefficient r is the following:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{(n - 1)(s_x s_y)}$$

In this formula, n is the number of observations, x_i is the value of the i^{th} observation of variable x , and y_i is the value of the i^{th} observation of variable y . $\bar{x}$ and $\bar{y}$ are the means of variables x and y , and s_x and s_y are the standard deviations of variables x and y . The Σ symbol indicates that the n values of the product $(x_i - \bar{x})(y_i - \bar{y})$ are to be added together.

INTERPRET THE DATA

- First, practice reading the data table. For the eighth observation ($i = 8$), what are x_i and y_i ? For which phylum are these data?
- Next, we'll calculate the mean and standard deviation for each variable (to review these topics, see the Scientific Skills Review in Appendix D). **(a)** The **mean** ($\bar{x}$) is the sum of the data values divided by n , the number of observations: $\bar{x} = \frac{\sum x_i}{n}$. Calculate the mean number of miRNAs ($\bar{x}$) and the mean number of cell types ($\bar{y}$) and enter them in the data table (to calculate $\bar{y}$, replace each

Data from the Study

Animal Phylum	i	No. of miRNAs (x_i)	$(x_i - \bar{x})$	$(x_i - \bar{x})^2$	No. of Cell Types (y_i)	$(y_i - \bar{y})$	$(y_i - \bar{y})^2$	$(x_i - \bar{x})(y_i - \bar{y})$
Porifera	1	5.8			25			
Platyhelminthes	2	35			30			
Cnidaria	3	2.5			34			
Nematoda	4	26			38			
Echinodermata	5	38.6			45			
Cephalochordata	6	33			68			
Arthropoda	7	59.1			73			
Urochordata	8	25			77			
Mollusca	9	50.8			83			
Annelida	10	58			94			
Vertebrata	11	147.5			172.5			
		$\bar{x} =$ $s_x =$		$\Sigma =$	$\bar{y} =$ $s_y =$		$\Sigma =$	$\Sigma =$

Data from B. Deline et al., Evolution of metazoan morphological disparity, *Proceedings of the National Academy of Sciences USA* 115:E8909–E8918 (2018).

x in the formula with a y). **(b)** Next, calculate $(x_i - \bar{x})$ and $(y_i - \bar{y})$ for each observation, recording your results in the appropriate column. Square each of those results to complete the $(x_i - \bar{x})^2$ and $(y_i - \bar{y})^2$ columns; sum the results for those columns. **(c)** Next, we'll calculate the **standard deviation**, s , which describes the variation found in a data set. Calculate s_x , the standard deviation of the miRNA data, using the following formula, where n = the number of observations. [Use the sum of $(x_i - \bar{x})^2$ you found in part (b).]

$$s_x = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

Repeat to find s_y by substituting y for x in this equation.

- Next, we'll calculate the correlation coefficient r for the variables x and y . **(a)** First, use the results from question 2(b) to complete the $(x_i - \bar{x})(y_i - \bar{y})$ column; sum the results in that column. **(b)** Now find the product of $s_x s_y$ using your results from 2(c) above. Plug these values into the formula for r .
- Do these data indicate that miRNA diversity and animal complexity are negatively correlated, positively correlated, or uncorrelated? Explain.
- What does your analysis suggest about the role of miRNA diversity in the evolution of animal complexity?

➔ **Instructors:** A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

as the top predators of the marine food web. By 450 million years ago, groups that diversified during the Cambrian period began to make an impact on land. Arthropods were the first animals to adapt to terrestrial habitats, as indicated by fragments of arthropod remains and by well-preserved fossils from several continents of millipedes, centipedes, and spiders. Another clue is seen in fossilized fern galls—enlarged cavities that fern plants form in response to stimulation by

resident insects, which then use the galls for protection. Fossils indicate that fern galls date back at least 302 million years, suggesting that insects and plants were influencing each other's evolution by that time.

Vertebrates colonized land around 365 million years ago and diversified into numerous terrestrial groups. Two of these survive today: the amphibians (such as frogs and salamanders) and the amniotes (reptiles, including birds, and